When Creative Destruction Becomes Contagion: Financial Innovation and the Architecture of Systemic Risk

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1. Introduction

On 6 May 2010, electronic markets briefly lost roughly one trillion dollars in value within thirty-six minutes before prices recovered almost as quickly (SEC and CFTC, 2010). Thirteen years later, depositors withdrew forty-two billion dollars from Silicon Valley Bank in a single day, at a rate exceeding one million dollars per second, with withdrawal pressure amplified by social media and venture-capital networks (Federal Reserve OIG, 2023). Both events revealed how technologies designed to reduce friction can also accelerate fragility.

The work recognised by the 2025 Nobel Prize in economics shows that creative destruction, the displacement of incumbents by innovators, is the engine of sustained growth (Aghion and Howitt, 1992; Mokyr, 2002). In standard growth theory, creative destruction reallocates resources from less productive incumbents to more productive entrants. Finance complicates this logic because financial institutions are interconnected through balance sheets, counterparty networks, and shared infrastructure. When a financial innovation displaces an incumbent process or institution, the 'destruction' phase can trigger contagion rather than orderly reallocation, converting private efficiency into public fragility.

This essay argues that innovative financial technologies reduce systemic risk when they remove wasteful frictions such as exclusion, opacity, slow settlement, and poor information, but amplify systemic risk when they eliminate stabilising frictions such as time delays that slow contagion, diversity of judgement across institutions, and regulatory visibility over critical points of failure. A friction is stabilising when its removal increases speed or correlation of loss propagation across the financial network; removing it shifts risk from the individual institution to the system. The net effect is conditional on institutional design, consistent with Mokyr's (2002) insight that sustained innovation requires appropriate institutional scaffolding to deliver welfare gains.

2. Systemic Risk and Endogenous Fragility

Systemic risk concerns the failure of the financial system to absorb shocks without disrupting vital services to the real economy, such as payments, credit, insurance, and settlement (Bank of England, 2025). It is not the sum of individual risks. It emerges through interconnectedness, leverage, and liquidity mismatch. Danielsson and Shin's (2003) concept of endogenous risk is central here: some risks are not external shocks but are generated and amplified by behaviour within the financial system itself. Financial innovation matters because it changes the speed, correlation, and topology of these internal feedback loops.

This is what separates financial creative destruction from industrial creative destruction. When Kodak was displaced by digital photography, its suppliers adjusted over years. When a financial institution or protocol fails, counterparties can fail simultaneously because they share collateral, liquidity pools, and infrastructure. Aghion and Howitt (1992) modelled incumbent displacement as a Poisson arrival process in which the old monopolist loses rents to the innovator. In finance, the displaced entity does not simply lose rents; its failure can propagate losses through the network, destroying value far beyond the original disruption.

3. How Financial Technologies Reduce Systemic Risk

FinTech can genuinely reduce fragility by removing wasteful frictions. Digital accounts and mobile money have broadened access to formal finance: the World Bank reports that 79 per cent of adults globally now hold an account at a financial institution or mobile-money provider, up from 51 per cent in 2011 (World Bank, 2025). Wider inclusion diversifies the financial base and reduces dependence on unregulated lenders. FinTech also improves information. The Bank for International Settlements argues that Big Tech lenders exploit data-network-activity loops to assess borrowers whose traditional credit files are thin, potentially reducing adverse selection and improving credit allocation (BIS, 2019). Settlement innovation offers further benefits: the BIS proposes that tokenised platforms anchored by central-bank reserves could reduce reconciliation delays and improve cross-border payments (BIS, 2025a).

Inclusion reduces fragility only when new credit is well underwritten. Speed reduces settlement risk only when infrastructure remains operational under stress. Transparency reduces opacity only when the information is decision-useful and acted upon. FinTech reduces systemic risk when it improves the plumbing of finance: broader access, better information, faster reconciliation. It does not reduce systemic risk merely because it is digital.

4. Channel 1: Speed Amplification

Digital finance is powerful in normal times because of its speed, scale and automation; under stress, those same features can become destabilising. In ordinary markets, algorithmic trading improves liquidity: Hendershott, Jones, and Menkveld (2011) find that it narrows spreads and reduces adverse selection, especially for large-capitalisation stocks. This is the removal of a wasteful friction, but liquidity in normal times is not the same as resilience under stress. Brunnermeier and Pedersen (2009) show that market liquidity and funding liquidity can reinforce each other in spirals where liquidity suddenly evaporates across assets. Liquidity supplied by fast agents may be abundant in appearance but fragile in commitment. The 2010 Flash Crash exposed this distinction: the SEC and CFTC found that more than 20,000 trades across over 300 securities were executed at prices more than 60 per cent away from their values moments before (SEC and CFTC, 2010), while Kirilenko et al. (2017) concluded that high-frequency traders did not cause the event but contributed to extraordinary volatility during it.

The compression of crisis timescales is especially visible in deposit runs. Washington Mutual lost roughly 16.7 billion dollars over ten days in September 2008. Silicon Valley Bank lost 42 billion dollars, a quarter of its total deposits, in approximately ten hours on 9 March 2023, with another 100 billion dollars queued for the following morning (Federal Reserve OIG, 2023). The Financial Stability Board confirms the pattern: the three fastest deposit runs in March 2023 had outflows of approximately 20–30 per cent of deposits per day, two to three times the highest peak daily outflow in its survey, and far above the historical average of roughly one per cent per day (FSB, 2024b). Mobile banking and social media did not create SVB's underlying fragility; that stemmed from interest-rate mismanagement. They did, however, transform the speed of the run, compressing what might once have unfolded over days into hours and sharply narrowing the window for management and regulators to respond. Faster payments are socially valuable, but faster panic is a source of systemic fragility.

5. Channel 2: Correlated Fragility

Artificial intelligence may improve individual credit assessment, but systemic risk concerns correlation and tail loss, not average predictive accuracy. Acharya et al. (2017) define systemic risk as an institution being undercapitalised precisely when the system as a whole is undercapitalised. Adrian and Brunnermeier's (2016) CoVaR similarly measures how distress at one institution relates to distress in the wider system. This implies a crucial point: a technology can improve individual risk management while worsening system-wide fragility if many institutions converge on the same models, data, vendors, or signals.

The Financial Stability Board identifies AI-related vulnerabilities, including third-party dependencies, service-provider concentration, model risk, and increased market correlations from common AI models and data sources (FSB, 2024a). The Bank of England warns that common weaknesses in AI models could cause many firms to misestimate, misprice, or misallocate credit simultaneously (Bank of England, 2025).

A simple model illustrates the mechanism. Suppose n lenders each have a loan-loss variance σ^2. If their model errors have pairwise correlation ρ, the variance of the average system loss is:

$$\mathrm{Var}(\bar{L}) = \sigma^2\left(\rho + \frac{1-\rho}{n}\right)$$

Figure 1 plots this expression for n = 10 and n = 100. With zero correlation and 100 lenders, diversification reduces system-loss variance to one per cent of individual variance. As ρ rises from 0.05 to 0.50, the system-loss variance factor jumps from 0.06 to 0.505. Systemic risk has increased by 8.5 times, even though each lender's standalone risk is unchanged. With only 10 lenders, the curve shifts upward, showing that market concentration magnifies the systemic impact of correlated model errors. This is the central danger of AI-driven finance: private accuracy can coexist with public fragility when institutions converge on the same data, vendors, and algorithmic logic.

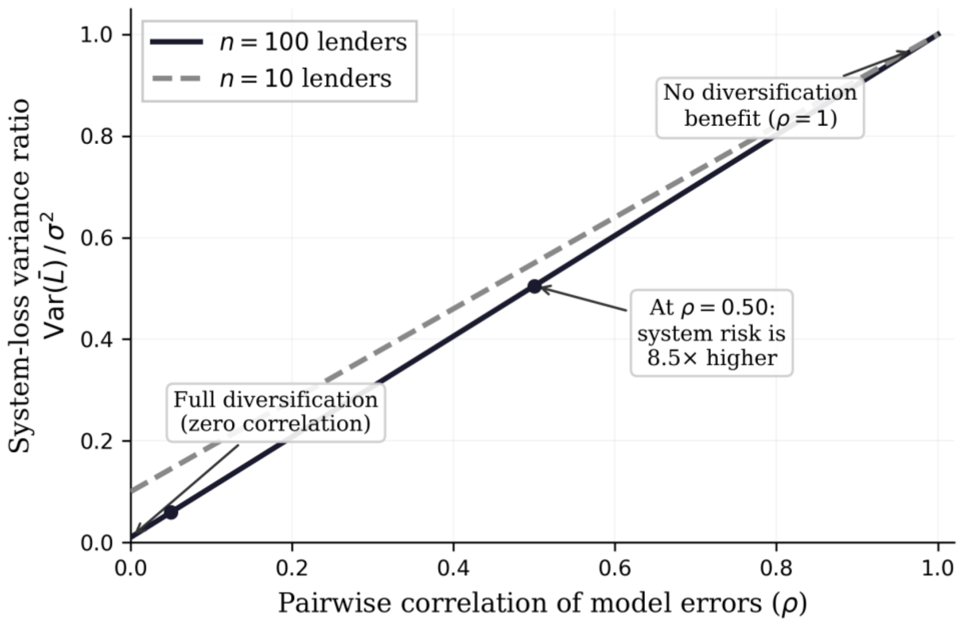

Figure 1. System-loss variance ratio as a function of pairwise model-error correlation (ρ), for 100 lenders and 10 lenders. As correlation rises, diversification benefits collapse: at ρ = 0.50, system risk is 8.5 times higher than at ρ = 0.05. Source: author's calculation.

6. Channel 3: Interconnected Opacity

Decentralised finance and stablecoins promise automatic settlement without traditional intermediaries, with transactions recorded on publicly visible blockchains. There is a serious argument that DeFi's radical transparency, where every transaction, collateral position, and liquidation is visible on-chain, represents a structural improvement over the opaque bilateral derivatives markets that amplified the 2008 financial crisis. If market participants can observe risk in real time, they should, in principle, be able to respond before it becomes systemic. This argument has merit, but visibility is not the same as resilience. A transparent ledger can reveal exposures, but it cannot guarantee liquid collateral, sound governance, or access to a lender of last resort. The collapse of the Terra ecosystem in May 2022 illustrates this starkly. TerraUSD (UST), an algorithmic stablecoin designed to maintain a dollar peg through arbitrage with its sister token LUNA, lost its peg on 7 May. Within seventy-two hours, approximately forty billion dollars in market capitalisation was destroyed (Federal Reserve, 2023). The Federal Reserve documented how the collapse propagated across multiple chains and assets: Terra's failure destabilised Celsius Network, whose insolvency triggered the collapse of Three Arrows Capital, whose losses contributed to the bankruptcy of FTX in November 2022. Each entity was connected through shared collateral, cross-platform liquidity, and DeFi protocol dependencies. The contagion was visible on-chain, but visibility did not prevent it.

DeFi also amplifies leverage in novel ways. Collateral can be reused across protocols, while automated liquidations can trigger cascading forced sales when asset prices fall, deepening the very decline that triggered them. In digital form, this mirrors the margin spirals described by Brunnermeier and Pedersen (2009). The European Systemic Risk Board reports that stablecoins had reached approximately 300 billion dollars by September 2025, with dollar-denominated stablecoins accounting for 99 per cent of stablecoin market capitalisation (ESRB, 2025). Economically, a reserve-backed stablecoin resembles a runnable money-like claim:

users expect redemption at par, but that promise depends on the quality, liquidity and transparency of the reserves. If redemptions occur at scale, issuers may be forced to sell those reserves, allowing stress in crypto markets to spill into traditional money markets.

Gai, Haldane, and Kapadia (2011) show that complexity and concentration in financial linkages can create tipping points where small shocks produce large systemic effects. FinTech shifts some systemic risk from regulated bank balance sheets toward platforms, protocols, and shared infrastructure where regulatory visibility is weaker and substitutability is lower. Transparency can reveal where risk concentrates, but it cannot, by itself, prevent that risk from spreading.

7. Policy Framework and Conclusion

The policy implication is not to resist innovation but to regulate by economic function and systemic footprint, not by technological label. A payment instrument, credit intermediary, or settlement platform should be judged by its functions, not whether it calls itself a bank, platform, protocol, or AI tool. The three channels identified above suggest specific responses. Speed amplification calls for adaptive circuit breakers and pre-positioned central bank liquidity facilities that can operate at the pace of digital bank runs. Correlated fragility calls for supervisory stress testing of ρ, the pairwise correlation of model errors across institutions, by mandating disclosure of which vendors, datasets, and algorithmic architectures firms share. Interconnected opacity calls for stablecoin reserve and redemption rules, resilience standards for critical third-party providers, and macroprudential oversight of how leverage and collateral move across DeFi protocols.

More broadly, five tests can guide the regulation of financial innovation. First, the function test: does the technology perform a core financial function such as payments, credit, or settlement? Second, the runability test: can users exit instantly at par? If so, liquidity buffers and stress tests are essential. Third, the correlation test: do many firms rely on the same model, dataset, or trading signal? Fourth, the concentration test: could one platform, cloud provider, or oracle become a single point of failure? Fifth, the observability test: can supervisors see the risk before stress? Crucially, because financial innovation is reflexive, static checklists succumb to Goodhart’s Law as market participants optimise around them. Mokyr’s 'institutional scaffolding' must therefore remain dynamic, evaluating not whether an innovation is technologically novel, but whether its failure topology is fast, correlated, or contagious.

The work recognised by the 2025 Nobel Prize in economics shows that creative destruction is the engine of economic growth, but also that its benefits depend on institutional preconditions (Aghion and Howitt, 1992; Mokyr, 2002). This essay has argued that financial creative destruction carries a distinctive danger: when the ‘destruction’ phase is contagious, it can undermine the very financial infrastructure on which future innovation depends. FinTech does not simply reduce or amplify systemic risk; it changes its architecture. Risk moves from branch queues to mobile apps, from human dealers to algorithms, from bank balance sheets to platforms, and from diverse credit judgement to common AI models. Its greatest contribution is removing wasteful friction. Its greatest danger is eliminating stabilising friction. The safest financial technologies are not those that make finance frictionless. They are those that remove wasteful frictions while preserving diversity, buffers and trust at machine speed.

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